

Saad Hassan

ASSISTANT PROFESSOR · TULANE UNIVERSITY

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Research Focus

My research spans accessible computing, human-computer interaction (HCI), and computational social science. I design and study human-AI systems that address accessibility at multiple levels: supporting communication with and among people with disabilities (e.g., sign language technologies and captioning systems), enabling access to digital information in educational and workplace contexts (e.g., reading support tools and accessible visualizations), and advancing accessibility compliance in the public sector (e.g., dashboards for tracking accessibility-barrier data). Across these areas, my work emphasizes “born-accessible,” personalizable design approaches that prioritize accessibility from the outset, rather than relying on post hoc fixes.

Education

Rochester Institute of Technology

PHD COMPUTING AND INFORMATION SCIENCE

Rochester, NY

August 2019 - May 2023

- Advisor: Matt Huenerfauth
- Thesis: Exploring Search-by-Video Technology for Searching Structured Human Movements: Insights From Sign Language Look-Up Systems.

Lahore University of Management Sciences

BS COMPUTER SCIENCE

Lahore, Pakistan

August 2015 - June 2019

Professional Experience

Tulane University

ASSISTANT PROFESSOR, *Full-time*

New Orleans, LA

July 2023 - present

- Department of Computer Science, School of Science and Engineering
- Director of the Accessibility (A11Y) Lab: <https://saadh.info/a11y/>
- Affiliated with the Center for Community-Engaged AI (CEAI): <https://sse.tulane.edu/cs/ceai>

Rochester Institute of Technology

GRADUATE RESEARCH ASSISTANT, *Part-time*

Rochester, NY

August 2019 - May 2023

- Center for Accessibility and Inclusion Research (CAIR): <https://cair.rit.edu/>
- Linguistics and Assistive Technology Lab (LATLAB): <https://latlab.ist.rit.edu/>

Google LLC

STUDENT RESEARCHER, *Part-time*

Remote (NY)

January 2023 - May 2023

- Sign Language Understanding Group (SLUG)
- Manager: Thad Starner

Meta Platforms, Inc.

STUDENT RESEARCHER, *Part-time*

Remote (NY)

September 2022 - Dec. 2022

- Audio Experiences Team at Meta Reality Labs (MRL)
- Mentors: Yao Ding, Christi Miller, and Agneya Kerure

Meta Platforms, Inc.

RESEARCH SCIENTIST INTERN, *Full-time*

Redmond, WA

May 2022 - September 2022

- Audio Experiences Team at Meta Reality Labs

Google LLCSTUDENT RESEARCHER, *Part-time*

- Google Perception

*Remote (NY)**September 2021 - May 2022***Google LLC**RESEARCH INTERN, *Full-time*

- Google Perception

*Remote (NY)**May 2021 - August 2021***Awards & Recognitions**

- 2025 **Best Paper Nomination (top 3.5% of 279 submissions)**, ASSETS 2025 (Publication FP.12)
- 2024 **Best Paper Honorable Mention (top 5% of 4028 submissions)**, CHI 2024 (Publication FP.7)
- 2024 **Best Paper Honorable Mention (top 5% of 4028 submissions)**, CHI 2024 (Publication FP.8)
- 2022 **Best Paper Nomination (top 7% of 132 submissions)**, ASSETS 2022 (Publication FP.4)
- 2022 **Duolingo Dissertation Research Award**, Recognized as one of six awardees for impactful dissertations on language learning technologies
- 2021 **Language Science and Computational Linguistics Student Excellence Award**, RIT, for outstanding research and coursework in Computational Linguistics

Grants & FundingExternal funding awarded: \$121,044; internal funding awarded: \$34,360; **total: \$155,404.****EXTERNAL FUNDING AWARDED**

2024–2026	Enhancing Expressive and Receptive ASL Learning using AI-Powered Tools , Saad Hassan (PI). Louisiana Board of Regents: RCS. Ranked 14/120	\$96,044
2025–2026	IUCRC Planning Grant: Tulane University (Partner Site): Center for Accessible Healthcare through AI-Augmented Decisions (AHEAD) , Aron Culotta (PI); Zizhan Zheng (Co-PI); Zhengming Ding (Co-PI); Sylvia Ley (Co-PI); Saad Hassan (Co-PI). National Science Foundation.	\$20,000
2022–2023	Design and Experimental Evaluation of Sign Language Look-up Tools , Saad Hassan. Duolingo Research Grant	\$5,000

INTERNAL FUNDING AWARDED

2025–2026	Public Dashboard for Accessibility Barrier Data , Saad Hassan (PI). Community-Engaged AI/DS Summer Research. Tulane University	\$8,500
2025–2026	AI-powered ASL Learning Platform for Emergency Medical Providers , Saad Hassan (PI). Carol Lavin Bernick Faculty Grant Program. Tulane University	\$10,000
2024–2025	Reading Support Tools for People who are Deaf or Hard of Hearing , Saad Hassan (PI). Committee on Research (COR) Fellowships. Tulane University	\$5,000
2024	In-situ Text Simplification Tools for Individuals with Low English Literacy , Saad Hassan (PI), Aaron Gershkovich (Student Researcher). CELT Award for Faculty Mentored Undergraduate Research	\$1,200
2024	Legal Document Reading Support for Native Chinese Speakers , Saad Hassan (PI), Oscar J Xiao (Student Researcher). CELT Award for Faculty Mentored Undergraduate Research	\$1,200
2024	Musical Instrument Digital Interface , Saad Hassan (PI), Madeline "Dylan" Murray (Student Researcher). CELT Award for Faculty Mentored Undergraduate Research	\$1,200
2023–2024	Understanding the Experiences and Technological Needs of Art Therapists , Saad Hassan (PI), Caitlin Chen (Student Researcher). CELT Award for Faculty Mentored Undergraduate Research	\$2,680
2023–2024	Art Appreciator for People with Autism , Saad Hassan (PI), Will Siver Wagman (Student Researcher). CELT Award for Faculty Mentored Undergraduate Research	\$1,580

TEACHING ENHANCEMENT GRANTS

2024 Faculty Innovation Grant , Saad Hassan (PI). Tulane University Innovation Institute (TUII)	\$2,500
2023 CELT Classroom Enhancement Grant , Saad Hassan. Tulane University	\$500

Publications

In computer science, publications in high-quality archival conference proceedings undergo rigorous peer review and are widely regarded as the primary measure of research impact. According to the Computing Research Association¹, conference publications are often preferred over journals, with premier conferences typically being more selective than journals.

I have published **33 works** in total, **16** of which were completed after joining Tulane (including **4** with Tulane students). These include **3 journal articles**, **12 peer-reviewed full-length papers**, **10 peer-reviewed short-length papers**, **2 peer-reviewed book chapters** (conference papers published as book chapters), **3 peer-reviewed workshop papers**, **2 peer-reviewed workshop proposals**, and **1 dissertation**. Summary of publication venues:

- **6 papers** at the **ACM Conference on Human Factors in Computing Systems (CHI, CORE A*)**² (5 full-length and 1 late-breaking work),
- **8 papers** at the **ACM SIGACCESS Conference on Computers and Accessibility (ASSETS, CORE A)** (2 full-length and 6 short-length papers),
- **4 papers** at the **International Web for All Conference (W4A, B1)**³ (3 full-length and 1 communication paper),
- **2 journal articles** in the **ACM Transactions on Accessible Computing (TACCESS, Norwegian Scientific Index 1)**⁴,
- **1 paper** at the **ACM International Conference on Mobile Human-Computer Interaction (MOBILEHCI, CORE A, published as a journal article)**,
- **1 paper each** at the **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR, CORE A*)**, the **Conference on Neural Information Processing Systems (NEURIPS, CORE A*)**, and the **Conference on Empirical Methods in Natural Language Processing Findings (EMNLP FINDINGS, CORE A)**.

Google Scholar Statistics (as of September 30, 2025): **286 citations, h-index 10, i10-index 12**.

Profile: <https://scholar.google.com/citations?user=oy6dZIIAAAAJ>

Authorship is marked in bold. Student authorship is **underlined and highlighted in green**. An asterisk (*) indicates equal contribution. Acceptance rates and total numbers of submitted/accepted papers are also provided when available.

MANUSCRIPTS UNDER REVIEW

[M.1] **Saad Hassan**, Motahhare Eslami, Nicholas Mattei, Aron Culotta, and John Zimmerman. AI In the Public Sphere: Citizen Engagement and Governance. (Submitted to The Handbook of Disruptive Urban Technologies and Society.)

PEER-REFEREED JOURNAL ARTICLES, PUBLISHED

- [J.3] **Saad Hassan**, Caluã de Lacerda Pataca, Akhter Al Amin, Laleh Nourian, Diego Navarro, Sooyeon Lee, Alexis Gordon, Matthew Watkins, Garreth W. Tigwell, and Matt Huenerfauth. 2024. Exploring the Benefits and Applications of Video-Span Selection and Search for Real-Time Support in Sign Language Video Comprehension among ASL Learners. *ACM Transactions on Accessible Computing* 17(3), Article 14, 35 pages. <https://doi.org/10.1145/3690647>
- [J.2] **Saad Hassan**, Abraham Glasser, Max Shengelia, Thad Starnier, Sean Forbes, Nathan Qualls, and Sam S. Sepah. 2023. Tap to Sign: Towards Using American Sign Language for Text Entry on Smartphones. *Proceedings of the ACM on Human-Computer Interaction* 7(MHCI), Article 227, 23 pages. <https://doi.org/10.1145/3604274>
- [J.1] **Saad Hassan**, Oliver Alonzo, Abraham Glasser, and Matt Huenerfauth. 2021. Effect of Sign-Recognition Performance on the Usability of Sign-Language Dictionary Search. *ACM Transactions on Accessible Computing* 14(4), Article 18, 33 pages. <https://doi.org/10.1145/3470650>

¹CRA: <https://cra.org/resources/best-practice-memos/>

²CORE Conference rankings: <https://portal.core.edu.au/conf-ranks/>

³ConferenceRanks listing: <http://www.conferenceranks.com>

⁴Norwegian Register of Scientific Journals: <https://kanalregister.hkdir.no/>

PEER-REFEREED FULL LENGTH PAPERS, PUBLISHED IN CONFERENCE PROCEEDINGS

- [FP.12] Oliver Alonzo* and **Saad Hassan***. 2025. A Review of 25 Years of Human–Computer Interaction Research on Reading Support Technologies for People with Disabilities Published in the ACM Digital Library. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. ACM, New York, NY, USA, 21 pages. <https://doi.org/10.1145/3663547.3746355> [Accepted, yet to be published] (83/279 = 29.7% accepted).
Conference Award: Best Paper Award Nomination (top 3.5%).
- [FP.11] **Saad Hassan**, Matyáš Boháček, **Chaelin Kim**, and Denise Crochet. 2025. Exploring the Design and Usage of an AI-Driven Video-Based American Sign Language Dictionary. [Accepted, yet to be published].
- [FP.10] Manfred Georg, Garrett Tanzer, Esha Uboweja, **Saad Hassan**, Maximus Shengelia, Sam Sepah, Sean Forbes, Thad Starner. 2025. FSboard: Over 3 Million Characters of ASL Fingerspelling Collected via Smartphones. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (CVPR '25)*. (2878/13,008 = 22.1% accepted).
Accepted papers: <https://cvpr.thecvf.com/Conferences/2025/AcceptedPapers>
- [FP.9] Caluã de Lacerda Pataca, **Saad Hassan**, Lloyd May, Michelle M. Olson, Toni D'Aurio, Roshan L. Peiris, and Matt Huenerfauth. 2025. Tactile Emotions: Multimodal Affective Captioning with Haptics Improves Narrative Engagement for d/Deaf and Hard-of-Hearing Viewers. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, Yokohama, Japan. ACM, 17 pages. <https://doi.org/10.1145/3706598.3713304> (1249/5020 = 25.1% accepted).
- [FP.8] **Saad Hassan**, Caluã de Lacerda Pataca, Laleh Nourian, Garreth W. Tigwell, Briana Davis, and **Will Zhenya Silver Wagman**. 2024. Designing and Evaluating an Advanced Dance Video Comprehension Tool with In-Situ Move Identification Capabilities. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, Article 888, 1–19. <https://doi.org/10.1145/3613904.3642710> (1060/4028 = 26.3% accepted).
Conference Award: Best Paper Honorable Mention (top 5%).
- [FP.7] Caluã de Lacerda Pataca, **Saad Hassan**, Nathan Tinker, Roshan Lalintha Peiris, and Matt Huenerfauth. 2024. Caption Royale: Exploring the Design Space of Affective Captions from the Perspective of Deaf and Hard-of-Hearing Individuals. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, Article 899, 1–17. <https://doi.org/10.1145/3613904.3642258> (1060/4028 = 26.3% accepted).
Conference Award: Best Paper Honorable Mention (top 5%).
- [FP.6] Thad Starner, Sean Forbes, Matthew So, David Martin, Rohit Sridhar, Gururaj Deshpande, Sam Sepah, Sahir Shahryar, Khushi Bhardwaj, Tyler Kwok, Daksh Sehgal, **Saad Hassan**, Bill Neubauer, Sofia Anandi Vempala, Alec Tan, Jocelyn Heath, Unnathi Utpal Kumar, Priyanka Vijayaraghavan Mosur, Tavenner M. Hall, Rajandeep Singh, Christopher Zhang Cui, Glenn Cameron, Sohier Dane, and Garrett Tanzer. 2023. PopSign ASL v1.0: An Isolated American Sign Language Dataset Collected via Smartphones. In *Proceedings of the 37th International Conference on Neural Information Processing Systems (NeurIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, Article 11, 184–196. <https://doi.org/10.5555/3666122.3666133> (322/985 = 32.7% accepted).
- [FP.5] Akhter Al Amin, **Saad Hassan**, Sooyeon Lee, and Matt Huenerfauth. 2023. Understanding How Deaf and Hard of Hearing Viewers Visually Explore Captioned Live TV News. In *Proceedings of the 20th International Web for All Conference (W4A '23)*. ACM, 54–65. <https://doi.org/10.1145/3587281.3587287> (16/32 = 50% accepted).
- [FP.4] **Saad Hassan**, Akhter Al Amin, Caluã de Lacerda Pataca, Diego Navarro, Alexis Gordon, Sooyeon Lee, and Matt Huenerfauth. 2022. Support in the Moment: Benefits and Use of Video-Span Selection and Search for Sign-Language Video Comprehension among ASL Learners. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '22)*. ACM, Article 29, 1–14. <https://doi.org/10.1145/3517428.3544883> (35/132 = 26.5% accepted).
Conference Award: Best Paper Nomination (top 7%).
- [FP.3] **Saad Hassan***, Akhter Al Amin*, Alexis Gordon, Sooyeon Lee, and Matt Huenerfauth. 2022. Design and Evaluation of Hybrid Search for American Sign Language to English Dictionaries: Making the Most of Imperfect Sign Recognition. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, Article 195, 1–13. <https://doi.org/10.1145/3491102.3501986> (324/2597 = 12.5% directly accepted).
- [FP.2] Akhter Al Amin*, **Saad Hassan***, Sooyeon Lee, and Matt Huenerfauth. 2022. Watch It, Don't Imagine It: Creating a Better Caption-Occlusion Metric by Collecting More Ecologically Valid Judgments from DHH Viewers. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, Article 459, 1–14. <https://doi.org/10.1145/3491102.3517681> (674/2597 = 25.9% accepted).
- [FP.1] Akhter Al Amin, **Saad Hassan**, and Matt Huenerfauth. 2021. Caption-Occlusion Severity Judgments across Live-Television Genres from Deaf and Hard-of-Hearing Viewers. In *Proceedings of the 18th International Web for All Conference (W4A '21)*. ACM, Article 26, 1–12. <https://doi.org/10.1145/3430263.3452429> (20/38 = 52.6% accepted).

PEER-REFEREED SHORT PAPERS, PUBLISHED IN CONFERENCE PROCEEDINGS

- [SP.10] **Nazmun Nahar Khanom, Aaron Gershkovich**, Oliver Alonzo, **Saad Hassan**. 2025. A Customizable AI-Powered Automatic Text Simplification Tool for Supporting In-Situ Text Comprehension. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. ACM, 5 pages. <https://doi.org/10.1145/3663547.3759713> (82/124 = 66.1% accepted).
- [SP.9] **Chaelin Kim, Cameron McLaren, Madhangi Krishnan, Nikhil Modayur, Saad Hassan**. 2025. Signing for Care: A Demo and Initial Evaluation of an American Sign Language Learning Tool for Emergency Medical Service Providers. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. ACM, 6 pages. <https://doi.org/10.1145/3663547.3759746> (82/124 = 66.1% accepted).
- [SP.8] Lloyd May, Alex Williams, **Saad Hassan**, Mark Cartwright, and Sooyeon Lee. 2024. Towards a Rich Format for Closed-Captioning. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*. ACM, Article 126, 1–5. <https://doi.org/10.1145/3663548.3688504> (58/87 = 66.7% accepted).
- [SP.7] LouAnne Boyd, Annuska Zolyomi, **Saad Hassan**, Seray B. Ibrahim, Guande Wu, and Kay Kender. 2024. Exploring the “Freedom to Be Me” through Design Sprints with Neurodivergent Scholars. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*. ACM, Article 96, 1–6. <https://doi.org/10.1145/3663548.3688527> (58/87 = 66.7% accepted).
- [SP.6] Matyáš Boháček and **Saad Hassan**. 2023. Sign Spotter: Design and Initial Evaluation of an Automatic Video-Based American Sign Language Dictionary System. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*. ACM, Article 92, 1–5. <https://doi.org/10.1145/3597638.3614497> (40/77 = 51.9% accepted).
- [SP.5] Akhter Al Amin, **Saad Hassan**, Matt Huenerfauth, and Cecilia Ovesdotter Alm. 2023. Modeling Word Importance in Conversational Transcripts: Toward Improved Live Captioning for Deaf and Hard of Hearing Viewers. In *Proceedings of the 20th International Web for All Conference (W4A '23)*. ACM, 79–83. <https://doi.org/10.1145/3587281.3587290> (16/32 = 50% accepted).
- [SP.4] **Saad Hassan**, Yao Ding, Agneya Abhimanyu Kerure, Christi Miller, John Burnett, Emily Biondo, and Brenden Gilbert. 2023. Exploring the Design Space of Automatically Generated Emotive Captions for Deaf or Hard of Hearing Users. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. ACM, Article 125, 1–10. <https://doi.org/10.1145/3544549.3585880> (330/968 = 34.1% accepted).
- [SP.3] **Saad Hassan**, Sooyeon Lee, Dimitris Metaxas, Carol Neidle, and Matt Huenerfauth. 2022. Understanding ASL Learners’ Preferences for a Sign Language Recording and Automatic Feedback System to Support Self-Study. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '22)*. ACM, Article 85, 1–5. <https://doi.org/10.1145/3517428.3550367> (43/73 = 58.9% accepted).
- [SP.2] **Saad Hassan**. 2022. Designing and Experimentally Evaluating a Video-Based American Sign Language Look-Up System. In *Proceedings of the 2022 Conference on Human Information Interaction and Retrieval (CHIIR '22)*. ACM, 383–386. <https://doi.org/10.1145/3498366.3505804>
Invited to the Doctoral Consortium.
- [SP.1] **Saad Hassan**, Matt Huenerfauth, Cecilia Ovesdotter Alm. 2021. “Unpacking the Interdependent Systems of Discrimination: Ableist Bias in NLP Systems through an Intersectional Lens.” In *Findings of the Association for Computational Linguistics: EMNLP 2021*. Association for Computational Linguistics, 3116–3123. <https://aclanthology.org/2021.findingsemnlp.267> (950/2549 = 34.9% accepted to main + findings).

BOOK CHAPTERS, PEER-REVIEWED PAPERS PUBLISHED AS BOOK CHAPTERS

- [C.2] Akhter Al Amin, **Saad Hassan**, Matt Huenerfauth. 2021. “Effect of Occlusion on Deaf and Hard of Hearing Users’ Perception of Captioned Video Quality.” In: Antona M., Stephanidis C. (eds) *Universal Access in Human-Computer Interaction: Access to Media, Learning and Assistive Environments*. HCII 2021. LNCS 12769. Springer, Cham. https://doi.org/10.1007/978-3-030-78095-1_16
- [C.1] **Saad Hassan**, Aiza Hasib, Suleman Shahid, Sana Asif, and Arsalan Khan. 2019. Kahaniyan — Designing for Acquisition of Urdu as a Second Language. In *Human-Computer Interaction – INTERACT 2019*, Paphos, Cyprus, Proceedings, Part II. Springer, 207–216. https://doi.org/10.1007/978-3-030-29384-0_13 (55/196 = 28% accepted).

PEER-REFEREED PAPERS, PUBLISHED IN WORKSHOP PROCEEDINGS

- [W.4] **Saad Hassan**, Matthew Seita, Larwan Berke, Yingli Tian, Elaine Gale, Sooyeon Lee, and Matt Huenerfauth. 2022. ASL-Homework-RGBD Dataset: An Annotated Dataset of 45 Fluent and Non-Fluent Signers Performing American Sign Language Homeworks. In *Proceedings of the LREC 2022 10th Workshop on the Representation and Processing of Sign Languages*. European Language Resources Association. <https://aclanthology.org/2022.signlang-1.11/>

- [W.3] Akhter Al Amin, **Saad Hassan**, Cecilia Ovesdotter Alm, Matt Huenerfauth. 2022. Using BERT Embeddings to Model Word Importance in Conversational Transcripts for Deaf and Hard of Hearing Users. In *The Second Workshop on Language Technology for Equality, Diversity and Inclusion (LT-EDI) at ACL 2022*. Association for Computational Linguistics, 35–40. <https://aclanthology.org/2022.ltedi-1.5>
- [W.2] **Saad Hassan**, Oliver Alonzo, Abraham Glasser, Matt Huenerfauth. 2020. Effect of Ranking and Precision of Results on Users' Satisfaction with Search-by-Video Sign-Language Dictionaries. In *Proceedings of the Sign Language Recognition, Translation & Production (SLRTP) Workshop at the 16th European Conference on Computer Vision (ECCV'20)*. Poster Presentation. (8/12 = 66.7% extended abstracts accepted)
- [W.1] **Saad Hassan**, Larwan Berke, Elahe Vahdani, Longlong Jing, Yingli Tian, Matt Huenerfauth. 2020. An Isolated-Signing RGBD Dataset of 100 American Sign Language Signs Produced by Fluent ASL Signers. In *Proceedings of the LREC 2020 9th Workshop on the Representation and Processing of Sign Languages*. European Language Resources Association, 89–94. <https://www.aclweb.org/anthology/2020.signlang-1.14>

WORKSHOP PROPOSALS

- [P.2] Lloyd May, **Saad Hassan**, Khang Dang, Sooyeon Lee, Oliver Alonzo. 2025. Participant Recruitment in Accessibility Research. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*. ACM, 7 pages. <https://doi.org/10.1145/3663547.3748639>
- [P.1] **Saad Hassan**, **Syeda Mah Noor Asad**, Motahhare Eslami, Nicholas Mattei, Aron Culotta, and John Zimmerman. 2024. PACE: Participatory AI for Community Engagement. *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing* 12(1), 151–154. <https://doi.org/10.1609/hcomp.v12i1.31610>

DEMONSTRATIONS AND EXHIBITIONS

- [D.3] **Nazmun Nahar Khanom**, **Saad Hassan**. 2024. **AI-Based Reading Support Tools for Deaf and Hard of Hearing People**. Tulane Research, Innovation, and Creativity Summit (TRICS), Tulane University, New Orleans, LA.
- [D.2] **Aaron Gershkovich** and **Saad Hassan**. 2023. **AI-Based Reading Support Tools**. Annual Undergraduate Poster Session, Tulane University, New Orleans, LA.
Award: Best Prototype
- [D.1] Allison Benz, Spencer Montan, Byron Behm, Nisha Cerame, Ikemefuna Chukwunyeremwa, Loam Shin, Jinlan Li, Tony Ellis, Thad Starnier, **Saad Hassan**, Abraham Glasser, Max Shengelia, Prerna Ravi, Sahir Shahryar, Colby Duke. 2022. **PopSign: Mobile Games to Teach Sign Language**. Imagine RIT 2022. Rochester, NY.

Teaching Experience and Curriculum Design

I have taught **three courses** at Tulane: **Human-Computer Interaction (CMPS 4330/6330)**, **Introduction to Data Science (CMPS 3160/6160)**, and **Research Seminar (CMPS 7010)**. The first two are cross-listed for **undergraduate** and **graduate** students, while the third is for CS PhD students. I also served as a **teaching assistant** for three courses during my studies. Across these offerings, I earned **strong evaluations**, averaging **4.8/5** on the “recommend instructor” question over the past three semesters. See the **table below** for details and School of Science and Engineering averages.

	HCI (Fall 2023)	HCI (Fall 2024)	Intro. to Data Science (Spring 2024)	Intro. to Data Science (Spring 2025)
1. Class sessions aided learning	4.46 (4.13)	4.81 ↑ (4.12)	4.75 (4.21)	4.56 (4.24)
2. Exams/projects matched objectives	4.38 (4.31)	4.83 ↑ (4.33)	4.75 (4.32)	4.72 (4.36)
3. Received meaningful feedback	4.23 (3.85)	4.67 ↑ (3.87)	4.46 (3.91)	4.48 (3.99)
4. Knowledge of subject increased	4.23 (4.28)	4.78 ↑ (4.29)	4.71 (4.33)	4.80 (4.36)
5. Recommend this course	4.00 (4.03)	4.56 ↑ (4.05)	4.79 (4.08)	4.76 (4.13)
6. Recommend this instructor	4.31 (4.20) ↑	4.83 (4.15)	4.91 ↑ (4.21)	4.84 (4.28)
7. Broadened diverse perspectives	4.00 (3.95)	4.67 ↑ (4.00)	4.48 (4.03)	4.28 (4.09)
8. Fostered inclusive space	4.85 (4.32)	4.67 (4.30)	4.74 (4.34)	4.72 (4.39)

INSTRUCTOR OF RECORD AT TULANE UNIVERSITY

Spring 2025: RESEARCH SEMINAR (CMPS 7010), Tulane University, *Course Instructor* (enrollment 13)

Introduces first-year Ph.D. students in Computer Science to research and teaching practices, including scholarly writing, literature synthesis, replication, and TA training.

Course website: <https://tulaneecs.github.io/cmeps7010/>

Spring 2025: INTRODUCTION TO DATA SCIENCE (CMPS 3160/6160), Tulane University, *Course Instructor* (enrollment 39)

Repeated offering of the course.

Course website: <https://nmattei.github.io/cmeps3160/>

Evaluations: *Recommend Instructor:* 4.84/5, *Recommend Course:* 4.76/5

Fall 2024: HUMAN-COMPUTER INTERACTION (CMPS 4660/6662), Tulane University, *Course Instructor* (enrollment 31)

Repeated offering of the course. **Supported by the Tulane Innovation Institute.**

Course website: <https://saadh.info/hci/>

Evaluations: *Recommend Instructor:* 4.83/5, *Recommend Course:* 4.56/5

Spring 2024: INTRODUCTION TO DATA SCIENCE (CMPS 3160/6160), Tulane University, *Course Instructor* (enrollment 39)

Inherited this course; students engage in the full data-science cycle and complete semester-long projects (Past projects: <https://nmattei.github.io/cmeps3160/projects/PastFinal/>).

Course website: <https://nmattei.github.io/cmeps3160/>

Evaluations: *Recommend Instructor:* 4.91/5, *Recommend Course:* 4.79/5

Fall 2023: HUMAN-COMPUTER INTERACTION (CMPS 4661/6661), Tulane University, *Course Instructor* (enrollment 15)

Developed Tulane's first HCI course. The theoretical component, modeled on MacKenzie's *Human-Computer Interaction: An Empirical Research Perspective*, covers six modules: perception and cognition, user research, experimental design, interaction modeling, evaluation methods, and human-AI interaction. The applied component is a semester-long project structured in two phases: *Getting the Right Design* (user research, qualitative analysis, personas, low-fidelity prototypes) and *Getting the Design Right* (high-fidelity prototyping, evaluations, iterative refinements, final testing).

Supported by a CELT classroom enhancement grant, the course included a guest lecture series featuring leading HCI and accessibility researchers from academia and industry.

Course website: <https://saadh.info/hci/>

Evaluations: *Recommend Instructor:* 4.31/5, *Recommend Course:* 4.0/5;

CELT Guest Lectures Evaluations: *Connecting to Classroom Experience:* 5/5, *Connecting to the Course:* 5/5

TEACHING ASSISTANTSHIPS

Fall 2022: Capstone in Human-Computer Interaction (HCIN 700), Rochester Institute of Technology, *Teaching Assistant*

Fall 2022: Research Methods in HCI (HCIN 600), Rochester Institute of Technology, *Teaching Assistant*

Spring 2019: Software Engineering (CS 360), Lahore University of Management Sciences, *Teaching Assistant*

Mentoring

At Tulane, I currently advise **4 PhD students**. Three are in their second year and one in her first. All of them have already published with me or have submissions under review. I have served on one PhD student's thesis prospectus, successfully defended in Fall 2024. Beyond PhD advising, I have supervised **2 master's students** in independent study or research positions and **6 undergraduate students** in funded positions, in addition to one undergraduate honors thesis. I have also advised **6 capstone teams** consisting of a total of **17 undergraduate students**. In total, I have advised **24 undergraduate students** as well as **one high school student** at Tulane. In my courses, two PhD students and two undergraduate students have contributed as teaching or class assistants. Some of my undergraduate advisees at Tulane have joined graduate programs at prestigious institutions, including Shayne Shelton (M.S. in Electrical and Computer Engineering, Purdue University), Miranda Diaz (M.S. in Information Science, Cornell University), Caitlin Chen (M.S. in Health Informatics, Boston University), and Will Silver Wagman (M.S. in Computer Science, Tulane University).

Before joining Tulane, and in collaborations outside the university, I mentored 23 students in research roles that resulted in several publications. My latest mentoring plan is available on my website in the "People" section: <https://saadh.info/>

PH.D. DISSERTATION ADVISEES

Areen Khalaila — Computer Science Ph.D. Student, Tulane University (2025–Present).

Co-advised with Rebecca Faust.

Chaelin Kim — Computer Science Ph.D. Student, Tulane University (2024–Present).

Co-author on published works (FP.11; SP.9).

Syeda Mah Noor Asad — Computer Science Ph.D. Student, Tulane University (2024–Present).

Co-author on published work (P.1).

Nazmun Nahar Khanom — Computer Science Ph.D. Student, Tulane University (2024–Present).

Co-author on published work (SP.10); Presenter on demo (D.3).

DISSERTATION COMMITTEES

Linsen Li — Computer Science Ph.D. Student, Tulane University. (2024–Present)

Thesis prospectus defended on November 20, 2024.

MASTER’S STUDENT RESEARCH ADVISEES

Justin Haysbert — Computer Science M.S. Student, Tulane University (2024–2025).

Independent study successfully completed in Spring 2025.

Will Silver Wagman — Computer Science M.S. Student (Psychology B.S.), Tulane University (2023–2024).

Recipient of CELT Undergraduate Sponsored Research Grant. Co-author on published work (FP.8). Best Paper Honorable Mention Awardee at CHI ’24.

UNDERGRADUATE STUDENT RESEARCH ADVISEES

Nikhil Modayur — Neuroscience & Computer Science B.S. Student, Tulane University (2024–Present).

Co-author on published work (SP.9).

Oscar Xiao — Finance & Computer Science B.S. Student, Tulane University (2024–Present).

Recipient of CELT Undergraduate Sponsored Research Grant.

Madhangi Krishnan — Neuroscience & Computer Science B.S. Student, Tulane University (2024–2025).

Co-author on published work (SP.9).

Cameron McLaren — Psychology & Computer Science B.S. Student, Tulane University (2024–2025).

Co-author on published work (SP.9).

Aaron Gershkovich — Biomedical Engineering & Computer Science B.S. Student, Tulane University (2024–2025).

Honorable Mention, CRA Outstanding Undergraduate Researcher Award; Recipient of CELT Undergraduate Sponsored Research Grant’ “Best Prototype” award winner at the 2024 Annual Undergraduate Poster Session, Tulane University; Co-author on published work (SP.10); Presenter on demo (D.2).

Michael M. Weild — Finance & Computer Science B.S. Student, Tulane University (2024).

Madeline ’Dylan’ Murray — Biology & Computer Science B.S. Student, Tulane University (2024).

Recipient of CELT Undergraduate Sponsored Research Grant.

Caitlin Chen — Neuroscience & Computer Science B.S. Student, Tulane University (2023–2024).

Recipient of CELT Undergraduate Sponsored Research Grant.

UNDERGRADUATE HONOR THESIS ADVISEES

Sydney Wade — Neuroscience & Computer Science B.S. Student, Tulane University (2024–2025).

Thesis successfully defended in Spring 2025.

UNDERGRADUATE CAPSTONE ADVISEES, MOST PROJECTS WERE RESEARCH ORIENTED

Term	Project	Students
2025-	<i>HealthVis</i> : Accessibility Health Visualizations	Emily Aymond, Madeline Davis, Jaimie Morris
2024-2025	American Sign Language Learning Tool for Emergency Medical Providers	Madhangi Krishnan, Cameron McLaren
	<i>Parla Playground</i> : Spanish Language Learning Game for Children	Miranda Diaz, Zoe S. Oboler, Sam DeMarinis
	Accessible Rich Caption Editor	Hubert Mendez, Ethan Schaferkotter, Izzy Blair
2023-2024	<i>CogniSync</i> : Brain-computer Interface for Wheelchair Navigation	Justin Haysbert, Ryan Stevens, Shayne Shelton, Gabriel Sagrera
	<i>College Compass</i> : Interactive Educational Path Visualization Tool	Devin Gutierrez, Tanner Martz

TEACHING AND CLASS ASSISTANTS

Cameron McLaren — Psychology & Computer Science B.S., Tulane.

CMPS 4661/6661 Human–Computer Interaction (Fall 2024).

CMPS 3160/6160 Introduction to Data Science (Spring 2025).

Amin Mir Fakhar — Computer Science Ph.D., Tulane.
CMPS 3160/6160 Introduction to Data Science (Spring 2025).

Lorraine Steigner — Classical Studies & Computer Science B.A., Tulane .
CMPS 4660/6662 Human-Computer Interaction (Fall 2024).
CMPS 3160/6160 Intro. to Data Science (Spring 2024).

Yunbei Zhang — Computer Science Ph.D., Tulane.
CMPS 3160/6160 Intro. to Data Science.

STUDENT ADVISEES PRIOR TO AND OUTSIDE TULANE

Lloyd May — Computer-based Music Technology & Acoustics Ph.D. Student, Stanford University (2023–Present).

Caluã de Lacerda Pataca — Computing & Information Sciences Ph.D. Student, RIT (2021–Present).
Co-author on published work (FP.9, FP.8, FP.4). Best Paper Award Nominee (ASSETS'22). Best Paper Honorable Mention (CHI'24).

Laleh Nourian — Computing & Information Sciences Ph.D. Student, RIT (2021–Present).
Co-author on published work (FP.7). Best Paper Honorable Mention (CHI'24).

Mathew Watkins — Human-Computer Interaction M.S. Student, RIT (2021–2022).
Co-author on published work (J.3).

Kira Hart — Secondary Education of DHH Students M.S. Student, RIT (2021–2022).

Abhay Dixit — Computer Science M.S. Student, RIT (2021–2022).

Sunnihih Manduva — Human-Computer Interaction M.S. Student, RIT (2020–2021).

Ruiwen Fan — Human-Computer Interaction M.S. Student, RIT (2019–2020).

Matyáš Boháček — Computer Science B.S. Student, Stanford University (2021–Present).
Co-author on published work (SP.6, FP.11).

Toni D'aurio — ASL-English Interpreting B.S. Student, RIT (2023–Present).
Co-author on published work (FP.9).

Max Shengelia — Web and Mobile Computing B.S. Student, RIT (2021–2023).
Co-author on published work (J.2, FP.10).

Nathan Tinker — ASL-English Interpreting B.S. Student, RIT (2022–2023).
Co-author on published work (FP.7).

Briana Davis — ASL-English Interpreting B.S. Student, RIT (2021–2023).
Co-author on published work (FP.8).

Diego Navarro — ASL-English Interpreting B.S. Student, RIT (2021–2022).
Co-author on published work (FP.4, J.3).

Alexis Gordon — Human-Centered Computing B.S. Student, RIT (2020–2022).
Co-author on published work (FP.3).

Megan Gross — ASL-English Interpreting B.S. Student, RIT (2021–2022).

Velvet Howland — ASL-English Interpreting B.S. Student, RIT (2021–2022).

Cole Inniss — Human-Centered Computing B.S. Student, RIT (2020–2021).

Taylor Harris — ASL-English Interpreting B.S. Student, RIT (2021–2022).

Jake Schwall — New Media Marketing B.S. Student, RIT (2021–2022).

Sarah Morgenthal — ASL-English Interpreting B.S. Student, RIT (2020–2022).

Konce Quispe — Computer Science B.S. Student, RIT (2019–2020).

Ben Leyer — ASL-English Interpreting B.S. Student, RIT (2019–2022).

Service Activities

At Tulane, I have served in service roles both within and outside the department across **four committees**. Outside the department, I have been a member of the **SSE Curriculum Committee** since Fall 2023, where I facilitated communication between the department and the committee, contributing to the approval of **10+ new CS courses** and **two new programs**.

Within the department, I have served as **Communications and Social Chair**, on the **Computer Science Curriculum Committee**, and on the **Undergraduate Studies Committee**.

Beyond Tulane, my service roles have included both conference leadership and reviewing activities. Since joining, I have served as **Workshop Chair** for ASSETS'23, **Proceedings Chair** for ASSETS'24, and as a **Best Paper Award Committee Member** for CHI'24 and CHI'25. I have also organized two workshops, including one on participant recruitment in accessibility research at ASSETS'25 and one on participatory AI for community engagement at HCOMP'24.

In reviewing roles, I have served on **two NSF panels**, as an **Associate Chair** at CHI for **four years**, on the ASSETS **Program Committee** for **four years**, and as an **Area Chair** at EMNLP. I have also served as a reviewer for the TACCESS journal for **three years**. Overall, I have reviewed **125+ submissions** for top HCI venues and **15+ submissions** for AI venues. I have received **three special recognitions** for outstanding reviewing.

WITHIN THE UNIVERSITY

CS Communications and Social Chair, Tulane University (2024–Present).

SSE Curriculum Committee, Member and CS Representative, Tulane University (2023–Present).

CS Curriculum Committee, Tulane University (2023–2024).

CS Undergraduate Studies Committee, Tulane University (2023–Present).

CONFERENCE LEADERSHIP ROLES

Best Paper Award Committee Member, CHI 2025, 2024.

Workshops Chair, ASSETS 2024.

Proceedings Chair, ASSETS 2023.

CONFERENCE PROGRAM COMMITTEE ROLES

Associate Chair, CHI (2023–2026).

Program Committee Member, ASSETS (2022–2025).

Area Chair, EMNLP 2023.

GRANT REVIEW PANELS

Proposal Reviewer and Panelist, National Science Foundation (NSF), Directorate for CISE, IIS (2024–2025).

Proposal Reviewer and Panelist, NSF SBIR/STTR Program (2024).

REVIEWING

Conference Reviewer

ACM Conference on Human Factors in Computing Systems (CHI) — Papers (2022–2026, **Special Recognition for Outstanding Reviews** in 2022 and 2023); Case Studies (2023); Late-Breaking Work (2023)

ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) — Posters (2024–2025)

ACM Conference on Designing Interactive Systems (DIS) — Papers (2025, **Special Recognition for Outstanding Reviews**)

ACM SIGACCESS Conference on Computers and Accessibility (ASSETS) — Technical Papers (2022–2025); Posters & Demos (2022–2025); Workshops (2022); Experience Reports (2022)

ACM Conference on Intelligent User Interfaces (IUI) — Posters & Demos (2023–2025)

IEEE Conference on Virtual Reality and 3D User Interfaces (VR) — Papers (2023)

International AAAI Conference on Web and Social Media (ICWSM) — Posters & Datasets (2023)

ACM SIGCHI Annual Symposium on Computer–Human Interaction in Play (CHI PLAY) — Work-in-Progress (2022)

ACM Symposium on Eye Tracking Research & Applications (ETRA) — Short Papers (2022)

ACM/IEEE International Conference on Human–Robot Interaction (HRI) — alt.HRI (2022)

ACM Interaction Design and Children Conference (IDC) — Work-in-Progress (2022)

ACM International Symposium on Wearable Computers (ISWC) — Notes & Briefs (2022)

Nordic Conference on Human–Computer Interaction (NORDICHI) — Papers & Critiques (2022)

Australian Conference on Human–Computer Interaction (OzCHI) — Case Studies & Late-Breaking Work (2021)

Journal Reviewer

ACM Transactions on Accessible Computing (TACCESS) (2022–Present)

ACM Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT) (2024–Present)

Ethics Reviewer

ACL Rolling Review (2024)

Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) (2024)

OTHER SERVICE ACTIVITIES PRIOR TO AND OUTSIDE TULANE

Communication Chair, Center for Accessibility and Inclusion Research, Rochester Institute of Technology (2020–2022).

Student Volunteer, Conference on Empirical Methods in Natural Language Processing (EMNLP 2021).

Student Volunteer, International Web for All Conference (W4A 2022).

Dissemination

I have delivered several research talks in courses, seminars, and centers at Tulane University, as well as at external venues. These exclude talks presented at conferences or workshops. In addition, some of my work has been featured in university news articles, and I have organized two workshops at academic conferences.

GUEST LECTURES IN UNIVERSITY COURSES

“Studying and Designing Human–AI Systems with and for People with Disabilities and Disability Communities,” *Center for Community Engaged AI, Tulane University* (September 2025); *Biomedical Engineering Seminar, Tulane University* (September 2025).

“Human–Computer Interaction and Accessibility Research,” *Introduction to Computer Science (CMPS 1500), Tulane University* (April 2025, November 2023); *Research Seminar (CMPS 7010), Tulane University* (November 2024); *CS for High School Students Outreach Program, Tulane University* (November 2024); *Graduate Orientation in Computer Science, Tulane University* (August 2023).

AI and Equity (Panel), *Tulane University, New Orleans, LA* (April 2024).

“Ubiquitous Computing for People with Disabilities,” *Human–Computer Interaction with Mobile, Wearable, and Ubiquitous Devices (HCIN 722), Rochester Institute of Technology* (April 2021).

INVITED TALKS AND PANELS

“Literature Review on HCI Research on Reading Support Technologies,” *Readability Research Consortium* (September 2023).

“AI-powered Linguistic Technologies for Deaf and Hard of Hearing Individuals (Panel),” *NOLA AI Conference, New Orleans, LA* (November 2023).

“AI-powered Sign Language Learning Tools,” *Duolingo Research, Pittsburgh, PA* (November 2023).

“Designing AI-based Applications to Benefit Deaf and Hard of Hearing Individuals and Sign Language Users: An HCI Perspective,” *Tulane University, New Orleans, LA* (March 2023); *University of Amsterdam, Netherlands* (February 2023); *Pennsylvania State University, State College, PA* (February 2023).

“Designing AI-based Applications to Benefit Deaf and Hard of Hearing Individuals and Sign Language Users,” *University of Alabama, Tuscaloosa, AL* (February 2023). *San Francisco State University, Remote* (February 2023).

“Designing and Experimentally Evaluating Applications of AI-based Technologies to Support Deaf and Hard of Hearing Individuals and Sign Language Users,” *California Polytechnic State University, San Luis Obispo, CA* (February 2023). *Seattle University, Seattle, WA* (January 2023). *Union College, Schenectady, NY* (January 2023). *Dickinson College, Carlisle, PA* (February 2023).

“Language and Speech-based Assistive Technologies for Deaf and Hard of Hearing Individuals,” *Johns Hopkins University*, (November 2022).

“Designing Emotive Captions for Deaf and Hard of Hearing Individuals,” *Meta Reality Labs, Redmond, WA* (August 2022).

“Sign Language-based Smartphone Interactions,” *Google People + AI Research (PAIR) Group, Remote* (April 2022).

“AI-based Technologies for Deaf and Hard of Hearing Individuals,” *Research Experience for Undergraduates (REU) Symposium, Remote* (July 2021).

WORKSHOPS ORGANIZED

Participatory AI for Community Engagement (PACE) Workshop, HCOMP 2024.
<https://tulaneecs.github.io/PACE/>

Participant Recruitment in Accessibility Research Workshop, ASSETS 2025.

<https://assets-recruitment-2025.github.io/>

MEDIA OUTREACH

“Saad Hassan: Researching at the Intersection of Computing and Accessibility,” *RIT News* (October 5, 2022).
www.rit.edu/news/researching-intersection-computing-and-accessibility

“Our 2022 Duolingo Research Grant Winners!” *Duolingo Blog* (September 16, 2022).
blog.duolingo.com/2022-duolingo-research-grant

Professional Development

CRA Career Mentoring Workshop, 2024

Participated in the Computing Research Association (CRA) Career Mentoring Workshop, engaging in small-group discussions with federal program managers and directors to develop long-term research strategies and lab management practices.

NCFDD Faculty Success Program, 2025

Enrolled in the National Center for Faculty Development & Diversity (NCFDD) Faculty Success Program, a semester-long intensive focused on research productivity, time management, and mentoring to support sustained academic growth.

TeachAccess Program, 2021

Completed training on accessible technology design practices and inclusive development methods to integrate accessibility principles into computing education and research.

Professional Memberships

Association for Computing Machinery (ACM): Special Interest Group on Accessible Computing (SIGACCESS), Special Interest Group on Human-Computer Interaction (SIGCHI), Special Interest Group on Information Retrieval SIGIR